

34. Graph these functions.

- (a) $g(t) = u(t) - u(t - 1)$
- (b) $g(t) = \text{rect}(t - 1/2)$
- (c) $g(t) = -4\text{ramp}(t)u(t - 2)$
- (d) $g(t) = \text{sgn}(t)\sin(2\pi t)$
- (e) $g(t) = 5e^{-t/4} u(t)$
- (f) $g(t) = \text{rect}(t)\cos(2\pi t)$
- (g) $g(t) = -6\text{rect}(t)\cos(3\pi t)$
- (h) $g(t) = \text{rect}(t)\text{tri}(t)$
- (i) $g(t) = \text{rect}(t)\text{tri}(t + 1/2)$
- (j) $g(t) = u(t + 1/2)\text{ramp}(1/2 - t)$
- (k) $g(t) = \text{tri}^2(t)$
- (l) $g(t) = \text{sinc}^2(t)$
- (m) $g(t) = |\text{sinc}(t)|$
- (n) $g(t) = \frac{d}{dt}(\text{tri}(t))$
- (o) $g(t) = \text{rect}(t + 1/2) - \text{rect}(t - 1/2)$
- (p) $g(t) = \left[\int_{-\infty}^t \delta(\tau + 1) - 2\delta(\tau) + \delta(\tau - 1) d\tau \right]$
- (q) $g(t) = 3\text{tri}(2t/3) + 3\text{rect}(t/3)$
- (r) $g(t) = 6\text{tri}(t/3)\text{rect}(t/3)$
- (s) $g(t) = 4\text{sinc}(2t)\text{sgn}(-t)$
- (t) $g(t) = 2\text{ramp}(t)\text{rect}((t - 1)/2)$
- (u) $g(t) = 4\text{tri}((t - 2)/2)u(2 - t)$
- (v) $g(t) = 3\text{rect}(t/4) - 6\text{rect}(t/2)$
- (w) $g(t) = 10\text{drcl}(t/4, 5)\text{rect}(t/8)$

36. A function $g(t)$ has the following description. It is zero for $t < -5$. It has a slope of -2 in the range $-5 < t < -2$. It has the shape of a sine wave of unit amplitude

and with a frequency of $1/4$ Hz plus a constant in the range $-2 < t < 2$. For $t > 2$ it decays exponentially toward zero with a time constant of 2 seconds. It is continuous everywhere.

- (a) Write an exact mathematical description of this function.
- (b) Graph $g(t)$ in the range $-10 < t < 10$.
- (c) Graph $g(2t)$ in the range $-10 < t < 10$.
- (d) Graph $2g(3 - t)$ in the range $-10 < t < 10$.
- (e) Graph $-2g((t + 1)/2)$ in the range $-10 < t < 10$.

38. The voltage illustrated in Figure E.38 occurs in an analog-to-digital converter. Write a mathematical description of it.

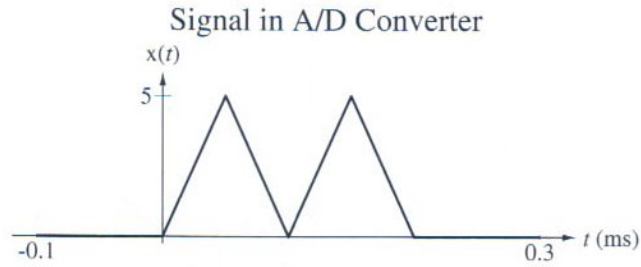


Figure E.38

Signal occurring in an A/D converter.

42. Describe the signal in figure E.42 two ways.

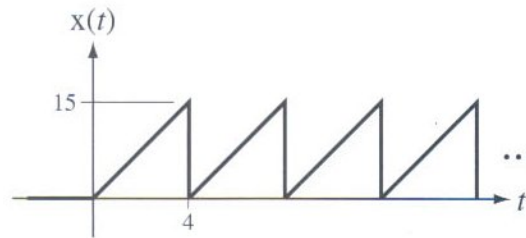
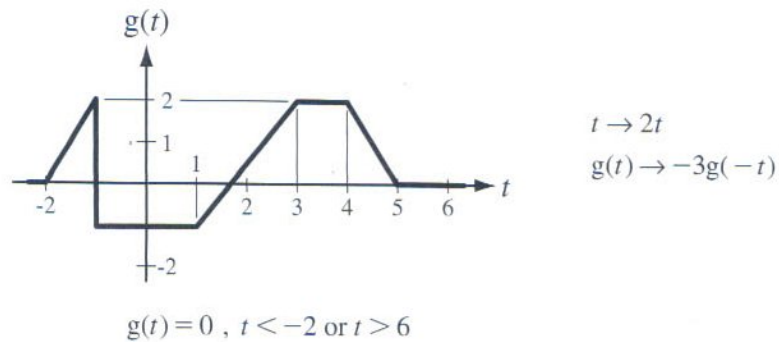


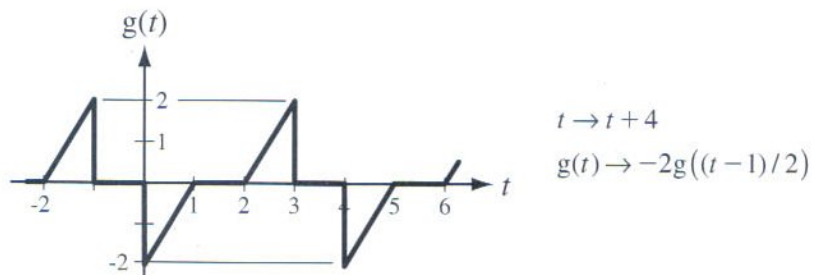
Figure E.42

- (a) As a ramp function minus a summation of step functions.
 (b) As a summation of products of triangle functions and rectangle functions.
45. Given the graphical definition of a function in figure E.45, graph the indicated shifted and/or scaled versions.

(a)



(b)



$g(t)$ is periodic with fundamental period, 4

58. Find and graph the even and odd parts of the function $x(t)$ in figure E.58.

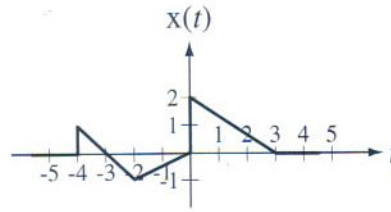


Figure E.58

Periodic Functions

59. For each of the following signals, decide whether it is periodic and, if it is, find the fundamental period.

- (a) $g(t) = 28\sin(400\pi t)$
- (b) $g(t) = 14 + 40\cos(60\pi t)$
- (c) $g(t) = 5t - 2\cos(5000\pi t)$
- (d) $g(t) = 28\sin(400\pi t) + 12\cos(500\pi t)$
- (e) $g(t) = 10\sin(5t) - 4\cos(7t)$
- (f) $g(t) = 4\sin(3t) + 3\sin(\sqrt{3}t)$

Signal Energy and Power

60. Find the signal energy of each of these signals.

- (a) $x(t) = 2\text{rect}(-t)$
- (b) $x(t) = \text{rect}(8t)$
- (c) $x(t) = 3\text{rect}(t/4)$
- (d) $x(t) = \text{tri}(2t)$
- (e) $x(t) = 3\text{tri}(t/4)$
- (f) $x(t) = 2\sin(200\pi t)$
- (g) $x(t) = \delta(t)$ (Hint: First find the signal energy of a signal which approaches an impulse in some limit, then take the limit.)
- (h) $x(t) = \frac{d}{dt}(\text{rect}(t))$
- (i) $x(t) = \int_{-\infty}^t \text{rect}(\lambda) d\lambda$
- (j) $x(t) = e^{(-1 - j8\pi)t} u(t)$